

Figure 3: Sine and cosine functions in Matplotlib

interval in the y-axis. If you look at the bottom of the image, you will see the option to zoom it. If you press the button and zoom along the y-axis, you will see the normal sine and cosine waves instead of straight lines. Better still, just comment out the line of code:

```
pt.plot(a,np.tan(a))

...as:

#pt.plot(a,np.tan(a))
```

...to obtain the program *plot2.py*. On execution of this program, you will get the output shown in Figure 3, in which only sine and cosine functions are plotted.

Sub-plots in Matplotlib

In the previous programs, multiple functions were plotted in a single image, but now we will try to place the plots of different functions into different sub-plots of the same image. Consider the program *plot3.py* given below to perform this task:

```
import matplotlib.pyplot as pt
import numpy as np

a = np.linspace(-10,10,1000)
pt.subplot(2,1,1)
pt.plot(a,np.sin(a))
pt.subplot(2,1,2)
pt.plot(a,np.tan(a))
pt.show()
```

There are only two lines of code that have been newly introduced in *plot3.py*. The line of code:

```
pt.subplot(2,1,1)
```

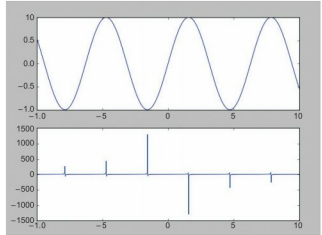


Figure 4: Sub-plots in Matplotlib

...shows that the plot has two rows, one column and the sub-plot should be placed in the first panel. The line of code:

```
pt.subplot(2,1,2)
```

...indicates that the sub-plot should be placed in the second panel of the plot with two rows and one column. The two functions plotted by the program are the trigonometric functions sine and tangent. On executing the program, *plot3.py* will produce the image shown in Figure 4, as the output.

Line styles and colours in Matplotlib

So far, we have only drawn lines in a single style without setting specific colours to specific lines in Matplotlib. But do not think that this is the limit of Matplotlib, which actually can do a lot of tasks with simple commands. In this section, we will look at how Matplotlib can use different styles and colours while drawing images. Consider the program *plot4.py* that uses different styles and colours while drawing the sine function, which we are already quite familiar with:

```
import matplotlib.pyplot as pt
import numpy as np
a = np.linspace(-10,10,1000)
pt.plot(a,np.sin(a-0),'o',color='red')
pt.plot(a,np.sin(a-1),'-',color='green')
pt.plot(a,np.sin(a-2),'--',color='blue')
pt.plot(a,np.sin(a-3),'.',color='yellow')
pt.plot(a,np.sin(a-4),'v',color='pink')
pt.plot(a,np.sin(a-5),'>',color='orange')
pt.show()
```

This program prints six sine waves such that each is one point apart from the nearby sine waves. The only line of code that requires an explanation is:

```
pt.plot(a,np.sin(a-0),'o',color='red')
```